

BUSINESS REQUIREMENTS DOCUMENT

NL Indian Bank Customer, Card & Transaction Behaviour Analysis

1. Executive Summary

NL Indian Bank wants to better understand its customers, how they use their cards, where they spend, which channels they prefer, and where fraud risk is highest. This document explains what the bank needs from a data analysis project that brings together **Customer, Card, Merchant and Transaction** information into one clear picture, shown through two dashboards.

The analysis covers **100,000 transactions**, **25,000 customers** (23,893 of whom made at least one transaction), **32,457 cards**, and **500 merchants** across **15 states** and **15 merchant categories**.

Headline Numbers

Metric	Value
Total amount spent	₹1.64B (₹1,642,467,223)
Total transactions	100,000
Successful transactions	91.15% (91,151)
Declined transactions	4.80% (4,796)
Failed transactions	4.05% (4,053)
Average spend per transaction	₹16,425
Customers who transacted	23,893
Transactions flagged for fraud	5,379 (5.38%)

The data shows that a small number of customer groups and spending categories drive most of the business, and that fraud risk is not spread evenly, it is much higher for certain card and channel combinations. These findings are explained in detail in the sections that follow, along with the changes the bank should consider making.

2. Project Background & Business Problem

Background

Banking in India today happens across many channels, cards, mobile apps, UPI, net banking, ATMs and physical branches, and across many types of merchants, from hospitals to e-commerce sites. As more customers join and use these channels, it becomes harder to see the full picture of how they behave simply by looking at one dataset at a time. NL Indian Bank already collects rich data about its customers, cards, transactions and merchants. But today these datasets sit separately, making it difficult to answer simple but important questions, such as which customers are most valuable, which channels carry the most fraud risk, and where the bank should focus its attention.

The Business Problem

The bank needs to bring these four datasets together and study them as one connected picture. Specifically, it wants to understand:

- Who its customers are and how their profile relates to their banking relationship.
- How customers use their cards, which types, networks and usage patterns are common.
- How customers behave when they transact, which channels, merchant categories and payment methods they use.
- Where transactions succeed, fail, or get declined, and where fraud is most likely to happen.

Without this connected view, the bank cannot easily spot patterns that would help it improve customer engagement, reduce fraud, or grow the business in the right areas.

3. Business Objectives

The main goal of this project is to give the bank a single, clear view of customer and transaction behaviour so it can make better decisions. The specific objectives are:

#	Objective	Why it matters (current data)
1	Understand which customers and segments drive the most value	Platinum and Gold customers already generate 68% of total spend
2	Reduce the number of failed and declined transactions	8.85% of transactions currently fail or are declined
3	Reduce fraud losses without upsetting genuine customers	5,379 transactions (5.38%) are currently flagged for fraud
4	Identify the merchant categories and channels worth investing in	Top 5 merchant categories generate roughly 65% of total spend
5	Improve the security of higher-risk channels	Some card and device combinations show fraud rates above 7%
6	Give business teams an easy way to see this data through dashboards	Two dashboards already built; this project defines how to run and improve them

4. Success Metrics & KPIs

These are the numbers the bank should track over time to see if this project is making a difference. Current values are shown as the starting point.

KPI	Current Value	Goal
Transaction success rate	91.15%	Increase over time
Declined transaction rate	4.80%	Decrease over time
Failed transaction rate	4.05%	Decrease over time
Fraud rate	5.38%	Decrease over time
Share of spend from Platinum & Gold customers	68.1%	Track and grow through better engagement
Share of spend from top 5 merchant categories	≈65%	Track for merchant strategy
Fraud from expired/blocked/lost cards	1,708 cases (31.8% of all fraud)	Reduce through card lifecycle automation

5. Hypothesis

The team tested four ideas against the data. All four were confirmed. These findings are the basis for the fraud-related requirements in this document.

#	Idea Tested	What the Data Showed	What the Bank Should Do
H1	Expired or blocked cards cause most preventable fraud.	Expired-card (1,010) and blocked-card (698) attempts make up almost 32% of all fraud flags.	Renew cards automatically before they expire and block payments on expired cards at the source.
H2	Fraud is higher on desktop/browser channels than on POS machines.	Credit card fraud on Windows PC reaches 7.06%, compared to 4.09% on POS terminals.	Add stronger identity checks (like step-up authentication) for browser-based payments.
H3	Platinum and Gold customers drive most spend but carry higher	Platinum (₹597.9M) and Gold (₹520.8M) customers generate 68.1% of all spend.	Add smart transaction limits for large purchases, without slowing down trusted customers.

#	Idea Tested	What the Data Showed	What the Bank Should Do
	fraud exposure on big transactions.		
H4	International transactions carry more fraud risk than domestic ones.	"Unusual Cross-Border Transaction" is the single biggest reason for international fraud, with 669 cases.	Send real-time alerts for international use and let customers turn international payments on or off in the app.

Why We Need to Do This Analysis

NL Indian Bank already has all the data it needs to answer important questions about its customers , but today that data sits in separate places, and nobody has connected the dots. This analysis matters because:

- **The bank is making decisions without the full picture.** Customer, card, merchant and transaction data are currently looked at separately, so it's hard to see how they connect. For example, the bank cannot easily tell whether its highest-spending customers are also its highest-risk customers, or whether certain merchant categories bring in a lot of revenue but also a lot of fraud.
- **Money is already being lost to fraud that follows clear patterns.** 5,379 transactions (5.38%) are currently flagged for fraud, and the patterns are not random , they cluster around specific card statuses, devices and transaction types. Without this analysis, the bank would keep losing money to problems that are actually predictable and preventable.
- **Not all customers and channels are equally valuable, but the bank cannot see this today.** Platinum and Gold customers already generate 68% of total spend, and just five merchant categories account for roughly 65% of spend. Without a connected view, the bank cannot prioritise where to focus service, marketing or risk controls.
- **Nearly 9% of transactions fail or get declined,** which affects customer experience every day. Right now, there is no clear, connected way to see why this happens or where it happens most, which makes it hard to fix.
- **Good data already exists, but it isn't being used to its full potential.** The bank has already cleaned and checked its data (see Section 8), but this effort only pays off if the data is then analysed and turned into clear, actionable insight , otherwise the work of collecting and cleaning it goes to waste.
- **Without this analysis, decisions are based on guesswork rather than evidence.** Bringing the four datasets together, as described in this document, gives the bank a factual, evidence-based foundation for improving customer engagement, reducing fraud, and growing the business in the right areas.

6. Stakeholder Analysis

The people and teams below have a stake in this project. Roles are based on common bank structures for this kind of work and should be confirmed with NL Indian Bank's own teams.

Stakeholder	Interest in the Project	Involvement
Bank Leadership / Sponsor	Overall results, business value and risk	Approves the project
Fraud & Risk Team	Fraud patterns and how to reduce them	Approves fraud-related changes
Cards & Payments Team	Card usage, card status and card-related fraud	Reviewed for input
Customer Experience Team	Customer segments and satisfaction	Reviewed for input
Merchant Relationship Team	Which merchant categories perform best	Reviewed for input
Data & Analytics Team	Builds and maintains the dashboards	Does the work
IT Team	Keeps the data flowing and secure	Does the work
Compliance / Audit	Making sure fraud reporting meets rules	Kept informed
Project Manager	Keeping the project on time and on scope	Runs the project

7. Functional Requirements

These are the things the dashboards and analysis must be able to do. Priority:
Must = required, Should = important, Could = nice to have.

ID	Requirement	Priority
FR-01	Show key numbers at a glance: total spend, total transactions, success rate, average spend, failed rate, number of customers, and fraud count.	Must
FR-02	Let users filter the data by year and account type.	Must
FR-03	Show customer spending on a map of India, broken down by state.	Should
FR-04	Show how much is spent through each payment channel (POS, Mobile App, Online Web, ATM).	Must
FR-05	Rank merchant categories by total spend so the bank can see which ones matter most.	Must
FR-06	Break down spending by customer segment (Basic, Standard, Gold, Platinum) and account type.	Must
FR-07	Show when transactions happen most , by day of week and hour of day.	Should
FR-08	Break down fraud reasons separately for domestic and international transactions.	Must
FR-09	Show which combination of payment method and device has the highest fraud rate.	Must
FR-10	Show how customer income relates to how much they spend.	Could
FR-11	Show what share of transactions succeed, get declined, or fail, by merchant risk level.	Must
FR-12	Let users move easily between the two dashboards.	Must
FR-13	Flag risky transaction patterns automatically, based on the four confirmed hypotheses in Section 12.	Must
FR-14	Keep a record of data quality checks so users can trust the numbers.	Should
FR-15	Allow the underlying tables to be exported for offline reporting.	Should

8. Non-Functional Requirements

These describe how the solution should behave, rather than what it should do.

Area	Requirement
Performance	Dashboards should respond quickly (a few seconds) even as more transactions are added.
Growth	The system should keep working well if the number of customers, cards or transactions grows well beyond current levels.
Data Quality	Known data issues should be flagged and tracked, not deleted, so nothing important is lost by mistake.
Privacy & Security	Personal details (name, income, occupation, age) should only be visible to people who need to see them.
Compliance	Fraud monitoring and reporting should follow the Reserve Bank of India's (RBI) rules on fraud risk management.
Audit Trail	Any rule changes based on this analysis should be logged and traceable.
Availability	Dashboards should be available during normal business hours, with an agreed uptime target.
Ease of Use	Anyone in the business should be able to read the dashboards without needing technical training.
Data Refresh	The data behind the dashboards should be updated on a regular, agreed schedule.

9. Data Requirements

The analysis depends on four connected tables. Each must stay accurate and linked correctly to the others.

Data Table	What it Contains	Size
Customers	Name, age, gender, income, occupation, segment, account type and how long they have banked with NL Indian Bank.	25,000 rows
Cards	Card type, network, credit limit, status (active/expired/blocked/lost), and issue/expiry dates.	32,457 rows
Merchants	Merchant name, category, location, risk level and rating.	500 rows
Transactions	Every transaction: amount, date, time, payment method, channel, device, status, and fraud flag.	100,000 rows

Data Quality Already Checked

The bank has already reviewed the data for common problems. The table below summarises what was found and how it was handled.

Issue Found	Rows Affected	How it was Handled
Misspelled value “Divorsed” in Marital Status	2,539	Corrected to “Divorced”
Unusually low yearly income (under ₹10,000)	95	Flagged for review, not deleted
Duplicate customer, card or merchant records	0	None found
Card status does not match its expiry date	12,621	Flagged for review, original status kept
Transaction happened before the customer joined the bank	16,440	Flagged for review, not deleted
Transactions with no fraud reason (expected, as most are legitimate)	94,621	Labelled “Not Flagged”

Note: No rows were deleted during cleaning. Unusual records were flagged for the business to review, so the original data is never lost.

10. Scope

In Scope

Analysis of the four core datasets: Customers, Cards, Merchants and Transactions.

Building and maintaining the two dashboards: Executive Overview and Risk, Fraud & Channel Analysis.

Reporting on customer segments, card usage, spending patterns, channel usage and fraud trends.

Documenting data quality issues found in the data and how they were handled.

Recommending business actions based on the four validated hypotheses (see Section 12).

Out of Scope

Building new source systems or changing how transactions are processed.

Real-time fraud blocking systems (this project identifies patterns; implementing live blocking rules is a separate initiative).

Data from outside the four provided tables, such as credit bureau scores or core banking ledgers.

Marketing campaign execution, even though the analysis may suggest target customer groups.

11. Assumptions & Constraints

Assumptions

The 23,893 “customers” shown on the dashboards means customers who made at least one transaction, not all 25,000 customers on the bank’s books. All amounts are in Indian Rupees (₹).
A “fraud flag” means a transaction was flagged by a rule for review , it does not always mean fraud was confirmed.
The dashboard’s year filter (2016–2026) reflects how long customers have banked with NL Indian Bank and how long cards have been issued, since actual transactions only run from January 2023 to August 2026.
Stakeholder names and reporting lines in Section 5 are assumed and should be confirmed.

Constraints

The analysis only covers the four tables provided; no external data (like credit bureau scores) was available.
Some transaction dates run slightly ahead of today’s date; this should be checked with the source system before the dashboards are used for regulatory reporting. The flagged records described in Section 8 (low income, mismatched card status, pre-join transactions) still need business review , they have not been resolved yet, only identified.

12. Risks

Risk	Likelihood	Impact	How to Reduce it
New fraud rules block genuine customers by mistake	Medium	High	Use step-by-step limits and a manual review step, especially for Platinum/Gold customers
Personal customer data is seen by people who don't need it	Medium	High	Restrict access based on role
Flagged data issues (Section 8) are not reviewed in time	Medium	Medium	Assign an owner and a deadline for each flagged issue
Dashboards slow down as more data is added	Medium	Medium	Test performance regularly as data grows
Teams disagree on priorities, causing delays	Medium	Medium	Agree scope and sign-off early with all stakeholders
Fraud reporting does not meet regulatory expectations	Low	High	Involve Compliance early and keep an audit trail

13. Data Sources

This project uses one Excel workbook (“NL_Banking.xlsx”) and two Tableau dashboards built from it.

Excel Workbook

Sheet	What it's For
Customers, Cards, Merchants, Transactions	The four core data tables described in Section 8.
Executive Summary, Key Metrics	Explains how each KPI is calculated.
Data Audit	Row counts and field details for every table.
Data Cleaning Log	Lists every data issue found and how it was handled (see Section 8).
Hypothesis Log	The four hypotheses and findings described in Section 12.
Pivot Tables 1–4	Ready-made summaries: spend by merchant category, fraud by channel, spend by segment, and fraud reasons.

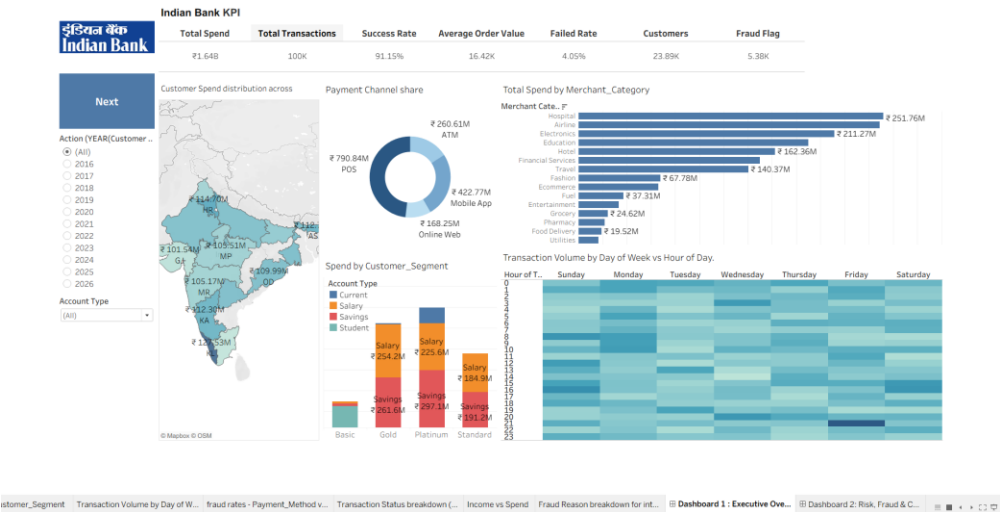
Dashboards

Two linked dashboards present this data. Each chart is shown individually in Sections 14 and 15.

14. Dashboard

1: Executive Overview, Customer Segmentation & Demographics

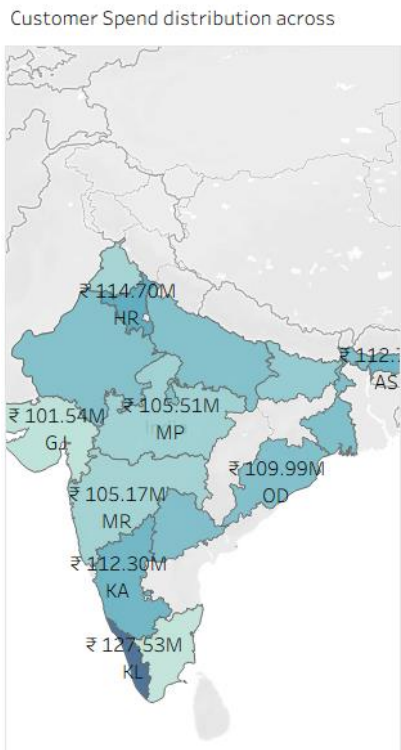
This dashboard gives a high-level view of overall performance, customer segments and where money is spent. It includes a KPI summary, a spend map, and four charts shown individually below.



Full view of Dashboard 1: Executive Overview.

15. Customer Spend Distribution Across States

This map shows how much customers in each state spend in total. Kerala (₹127.53M), Haryana (₹114.70M) and Assam (₹112.79M) are among the highest-spending states, while Tamil Nadu and Gujarat are on the lower end. pending is fairly



evenly spread across India's 15 states, with no single state dominating.
Chart: Customer spend distribution across states.

16. Payment Channel Share

This chart shows how much is spent through each channel. POS (in-store card payments) leads with ₹790.84M, followed by Mobile App at ₹422.77M, ATM at ₹260.61M, and Online Web at ₹168.25M. In-store card payments remain the bank's most-used channel by value.

Payment Channel share

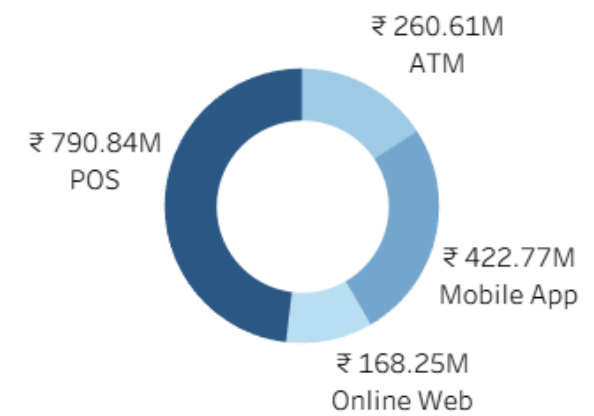


Chart: Payment channel share by value.

17. Spend by Customer Segment

This chart shows spend by customer segment (Basic, Standard, Gold, Platinum) and by account type (Current, Salary, Savings, Student). Platinum and Gold customers spend the most, mainly through Savings and Salary accounts, confirming that premium customers are the bank's most valuable group.

Spend by Customer_Segment

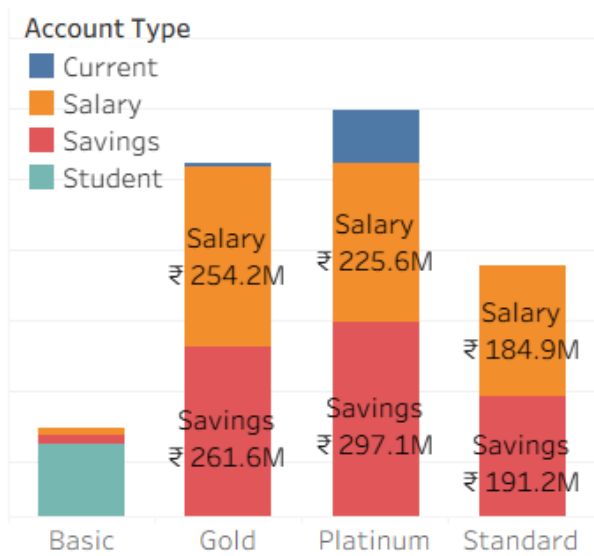


Chart: Spend by customer segment and account type.

18. Transaction Volume by Day of Week vs Hour of Day

This heatmap shows when customers transact most. Activity is fairly steady through the week, with some busier evening hours on Friday and mid-morning peaks on Monday and Tuesday. This can help the bank plan support staffing and fraud monitoring around busier hours.

Transaction Volume by Day of Week vs Hour of Day.

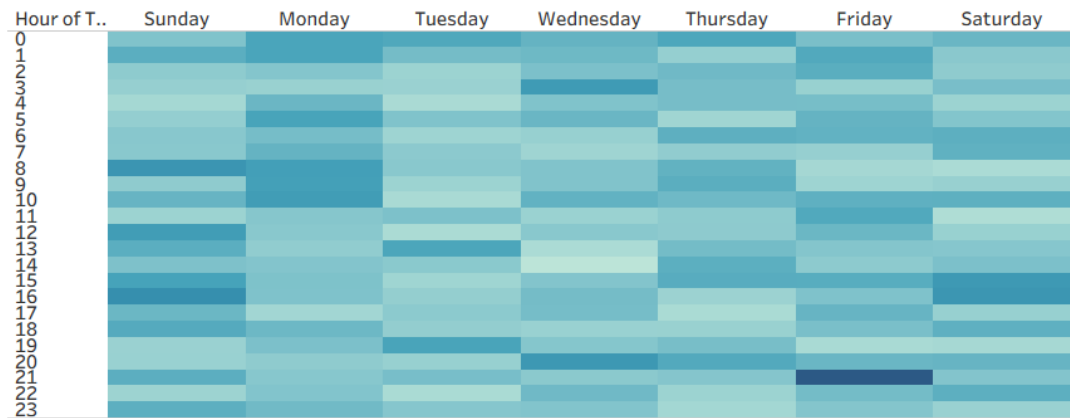


Chart: Transaction volume by day of week and hour of day.

19. Dashboard 2: Risk, Fraud & Channel Analysis

This dashboard focuses on risk and fraud. It shows where fraud happens most, how transactions succeed or fail, and how income relates to spending. Each chart is shown individually below.



Full view of Dashboard 2: Risk, Fraud & Channel Analysis.

20. Fraud Rates , Payment Method vs Device Type

This table shows the fraud rate for each combination of payment method and device, as currently displayed on the dashboard. The highest rate shown is UPI on Mac Desktop (6.14%), followed by UPI on Windows PC (5.42%) and Net Banking on iOS Mobile (5.27%). Desktop and browser-based access tends to carry a higher fraud rate than mobile apps or POS terminals across most payment methods.

fraud rates - Payment_Method vs Device_Type

Payme..	Android Mo..	ATM Machi..	iOS Mobile	Mac Desktop	POS Termin..	Windows PC
UPI	3.67%		4.23%	6.14%		5.42%
Credit Card	3.96%		4.31%	4.12%	4.09%	3.78%
Debit Card	4.26%	3.87%	4.07%	3.86%	4.02%	4.72%
Net Banking	3.06%		5.27%	2.89%		4.24%

Chart: Fraud rate by payment method and device type.

Note: The workbook’s own Pivot Table 2 (calculated directly from the latest transaction data) shows higher rates for some of the same combinations , for example, UPI on Mac Desktop at 7.72% and Credit Card on Windows PC at 7.06%. This dashboard image should be refreshed against the current data extract so the two stay in sync.

21. Transaction Status Breakdown (Successful vs Declined vs Failed)

This chart shows the share of successful, declined and failed transactions by merchant risk level. The pattern is almost identical across Low, Medium and High risk merchants , around 91% succeed, regardless of merchant risk level. This suggests merchant risk level alone does not explain most declines or failures.

Transaction Status breakdown (Successful vs Declined vs Failed)

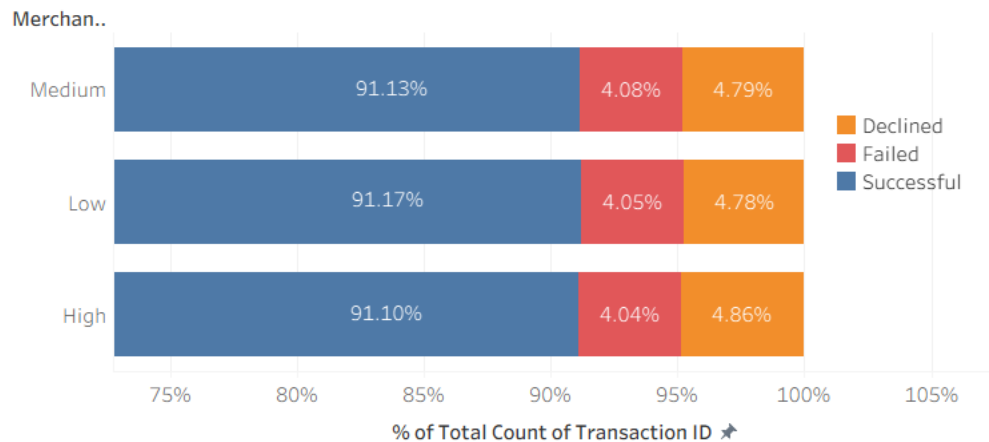


Chart: Transaction status by merchant risk level.

22. Income vs Spend

This chart compares customer income to how much they spend per transaction, coloured by segment. Basic and Standard customers (blue and teal) cluster at lower income and spend levels, while Platinum customers (red) spread across a much wider income range and include the highest individual transaction amounts.

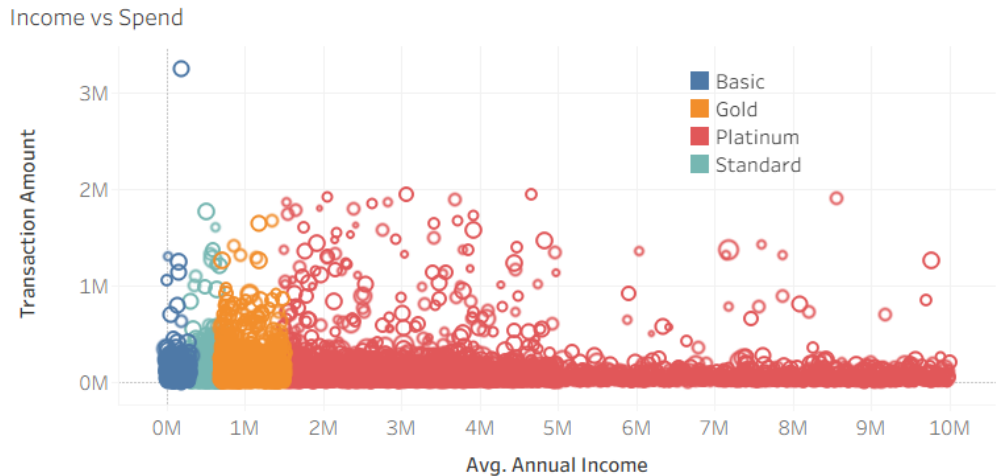


Chart: Income vs spend by customer segment.

23. Fraud Reason Breakdown for International vs Domestic Transactions

This chart splits fraud reasons by domestic and international transactions. Domestic fraud is mostly caused by expired cards (963 cases), high-risk merchant activity (691) and rapid repeated attempts (678). International fraud is almost entirely caused by one reason: unusual cross-border activity (669 cases). This confirms the bank needs different fraud rules for domestic and international transactions.

Fraud Reason breakdown for international vs domestic transactions.

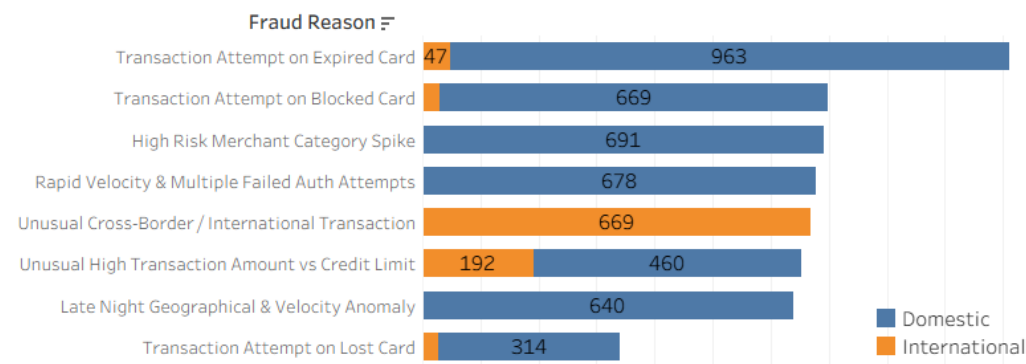


Chart: Fraud reasons, domestic vs international.

24. Data Dictionary Summary

A simple summary of the key fields used in this analysis.

Table	Field	What it Means
Customers	Customer_Segment	Basic, Standard, Gold or Platinum , the bank's customer tier
Customers	Annual_Income	Customer's yearly income
Customers	Customer_Since	Date the customer joined the bank
Cards	Card_Status	Active, Expired, Blocked or Lost
Cards	Card_Network	Visa, Mastercard, RuPay or American Express
Cards	Credit_Limit	Maximum amount the card can spend
Merchants	Merchant_Category	Type of business (e.g., Hospital, Travel, Fashion)
Merchants	Merchant_Risk_Level	Low, Medium or High risk rating for the merchant
Transactions	Transaction_Amount	How much the transaction was for
Transactions	Transaction_Status	Successful, Declined or Failed
Transactions	Fraud_Flag	1 if the transaction was flagged for review, 0 if not
Transactions	Fraud_Reason	Why a transaction was flagged (blank if not flagged)
Transactions	Is_International	Whether the transaction happened outside India

This BRD should be read together with **NL_Banking.xlsx** and the two dashboards referenced in Section 13.